Homework: Do spectral gradient updates help?

Optimization for Machine Learning – Teacher: Tam Le

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1 Task

Answer the question:

When, and in what sense, do spectral gradient updates improve gradient-based optimization and machine learning?

Investigate the question through the various optimization and machine learning concepts seen in class such as conditioning, momentum and empirical risk minimization. This list is suggestive, not exhaustive. For instance, a starting point can be to compare gradient method and spectral gradient method on a standard objective function. You will be assessed on the relevance and quality of your investigation, not on the number of experiments.

Choice of models and objective functions. Base your study on one or more setting. Some examples include

– linear regression;
– a small neural network;
– matrix factorization.

This list is suggestive, not exhaustive. Feel free to propose another relevant example seen in class, or from the machine learning literature. Toy (and reproducible) models are highly welcome; large-scale experiments will not receive additional credit simply because of their scale.

Implementation. Use `PyTorch` to implement the models, optimization methods, and numerical experiments. In particular, use automatic differentiation to compute gradients. You are encouraged to use built-in optimizers whenever they are available.

Use of generative AI. Generative AI (LLMs) may be used for coding, analysis, visualization, and writing. You remain responsible for the relevance of the experiments and for every submitted result and claim. Briefly state how AI was used (up to 5 lines) in a paragraph at the end. If any idea came from an LLM, state it explicitly. You can also use open source code available on the internet.

Discussions with other students. You may discuss about the homework with other students: implementation, experimental choices etc ... Each student must nevertheless define and conduct their own investigation, make their own experimental choices, and submit an independently written report and code. You remain responsible for checking all code used and for the validity of your results and conclusions.

Connections with the literature. You may connect the ideas investigated in your project with **relevant** literature beyond the material covered in class. Explain how selected references inform your questions, experimental choices, or interpretation of the results. Meaningful connections with the literature will be appreciated, but they are not mandatory: a strong connection with the course content is preferable to superficial or wrong use of external references.

2 Assessment

Midpoint progress report. Approximately halfway through the homework period (see the deadline indicated on Moodle), submit a preliminary version of your code and a progress report of at most one page containing:

1. the work completed so far;
2. any difficulties or uncertainties encountered;
3. a summary of your discussions with other students and your reflections on them;
4. a self-assessment, together with the experiments and decisions planned for the remainder of the project.

This midpoint report is intended to encourage early experimentation and to document how the project develops over time.

Final submission. Submit a report of **at most four pages**, including figures and tables. You can do it in French or English. Submit the code as a python file and/or a jupyter notebook. This means the main report should normally be approximately three pages with potential overlap on the fourth page. This limit of pages includes references and the brief AI-use statement.

Assessment criteria. The homework is graded out of 20 points:

Midpoint progress report	5 points
Redaction and clarity	3 points
Overall quality of the report	3 points
Interpretation and discussion	3 points
Experiments	3 points
Code and technical validity	3 points

The code and technical validity is assigned approximately as follows:

– 0: You did not use pytorch or incorrect code was not checked and some algorithms are wrong;
– 1: a substantial error affects part of the conclusions;
– 2: the implementation appears mostly correct but I could not check it easily;
– 3: a clear presentation effort makes correctness easy to check; the experiments are readily reproducible with a well structured source code and a notebook.